

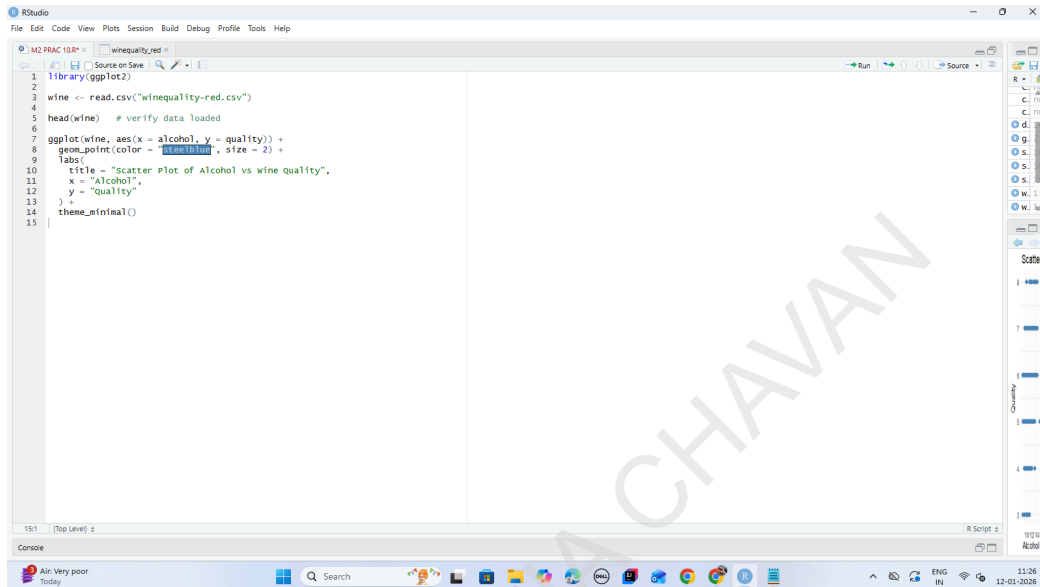
SHETH L.U.J. AND SIR M.V. COLLEGE
SUBJECT: DATA ANALYSIS WITH R

PRACTICAL NO. 10 - 12

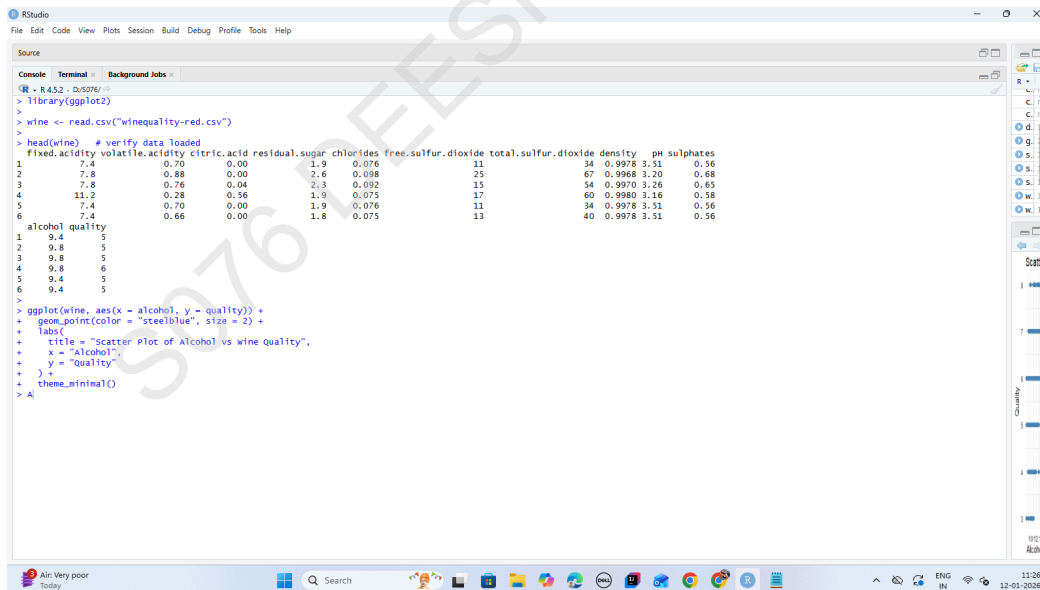
AIM:

10 Creating graphical reports using ,ggplot2 (R).

- Scatter Plots



```
1 library(ggplot2)
2 wine <- read.csv("winequality-red.csv")
3
4 head(wine) # verify data loaded
5
6
7 ggplot(wine, aes(x = alcohol, y = quality)) +
8   geom_point(color = "steelblue", size = 2) +
9   labs(
10     title = "Scatter Plot of Alcohol vs wine quality",
11     x = "Alcohol",
12     y = "Quality"
13   ) +
14   theme_minimal()
15
```



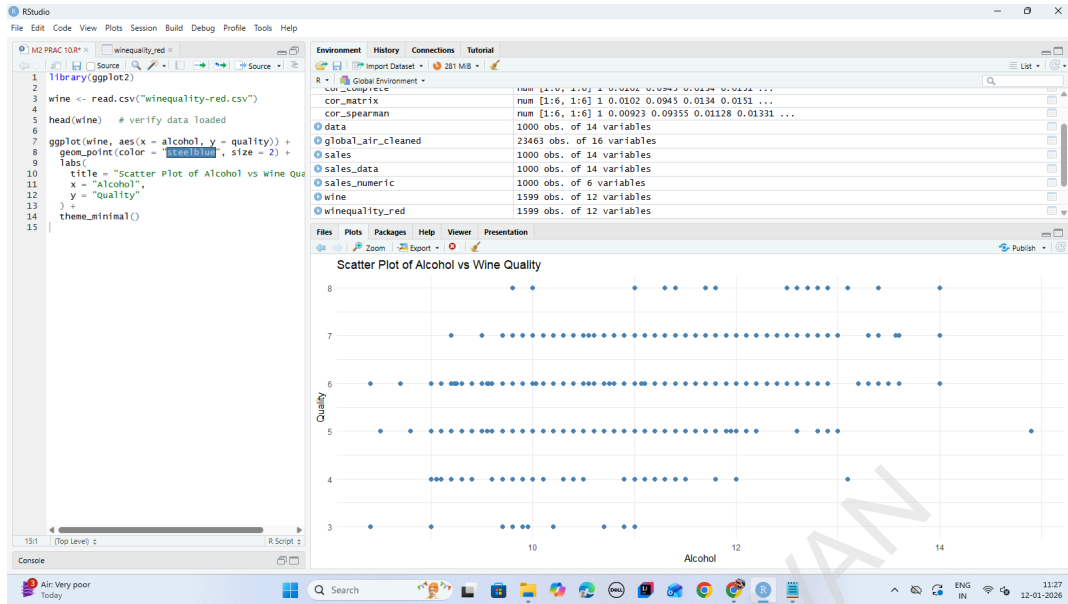
```
> library(ggplot2)
> wine <- read.csv("winequality-red.csv")
> head(wine) # verify data loaded
  fixed.acidity volatile.acidity citric.acid residual.sugar chlorides free.sulfur.dioxide total.sulfur.dioxide density  pH sulphates
1      7.4          0.70       0.00        1.9      0.076         11      34 0.9978 3.51 0.56
2      7.8          0.88       0.00        2.6      0.098         25      67 0.9968 3.20 0.68
3      7.8          0.76       0.04        2.3      0.092         15      54 0.9970 3.26 0.65
4     11.2          0.28       0.56        1.0      0.075         17      60 0.9980 3.16 0.58
5      7.4          0.70       0.00        1.9      0.076         11      34 0.9978 3.51 0.56
6      7.4          0.66       0.00        1.8      0.075         13      40 0.9978 3.51 0.56

alcohol quality
1    9.4      5
2    9.8      5
3    9.8      5
4    9.8      6
5    9.4      5
6    9.4      5

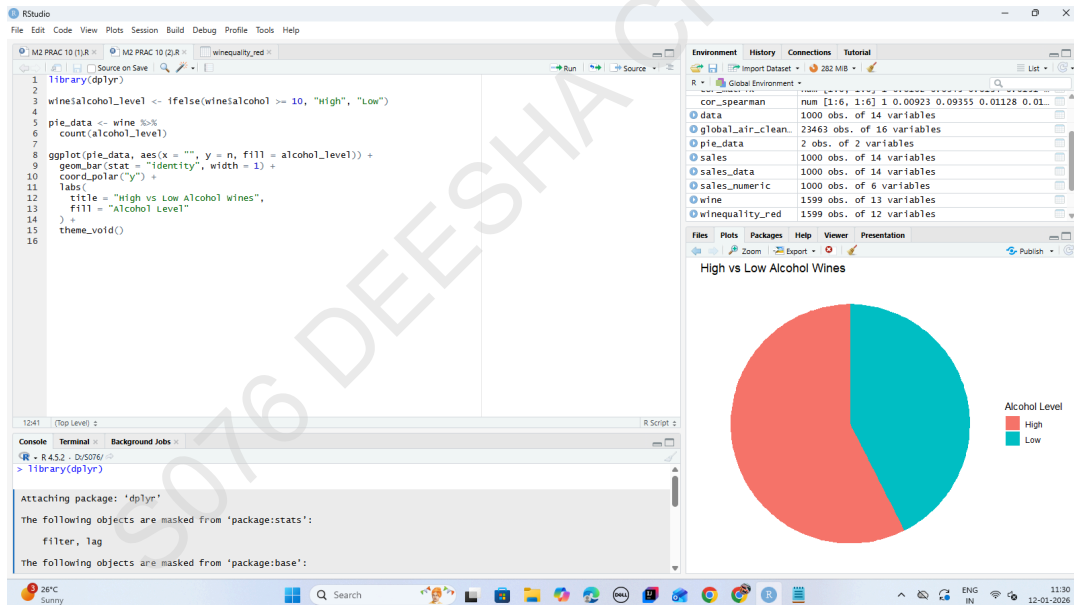
> ggplot(wine, aes(x = alcohol, y = quality)) +
+   geom_point(color = "steelblue", size = 2) +
+   labs(
+     title = "Scatter Plot of Alcohol vs wine quality",
+     x = "Alcohol",
+     y = "Quality"
+   ) +
+   theme_minimal()
> A
```

NAME : DEESHA CHAVAN
ROLL NO.: S076

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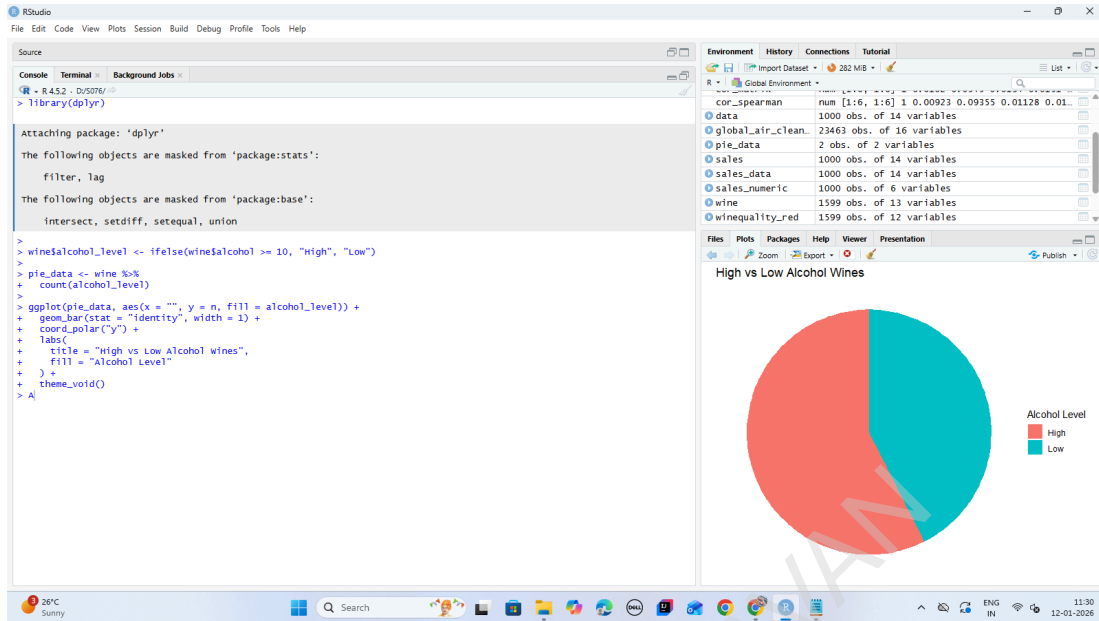
- Pie Charts



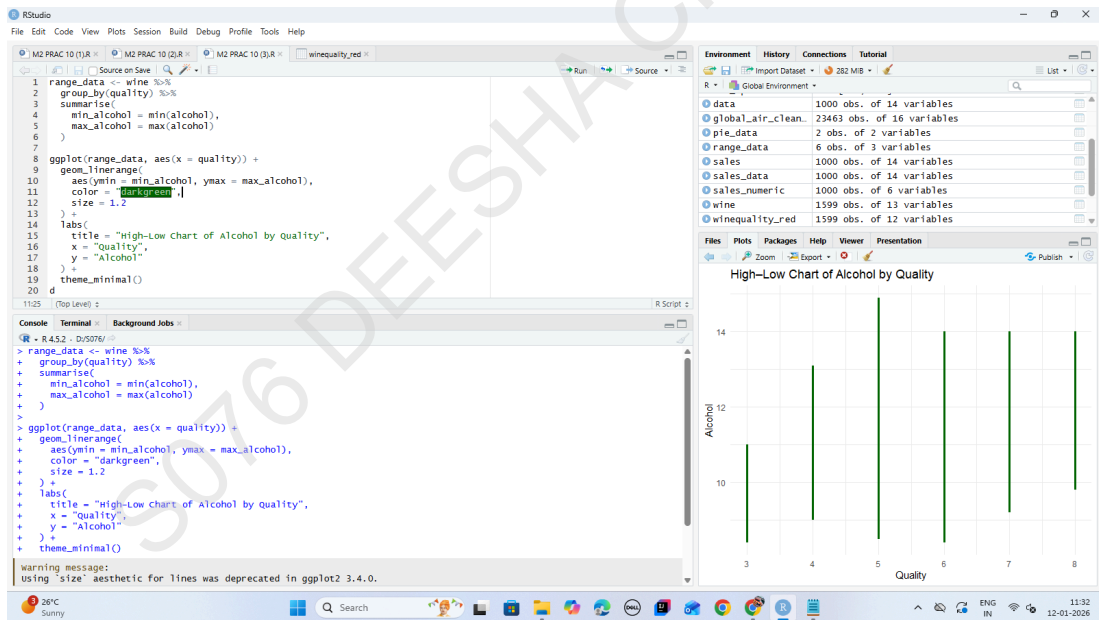
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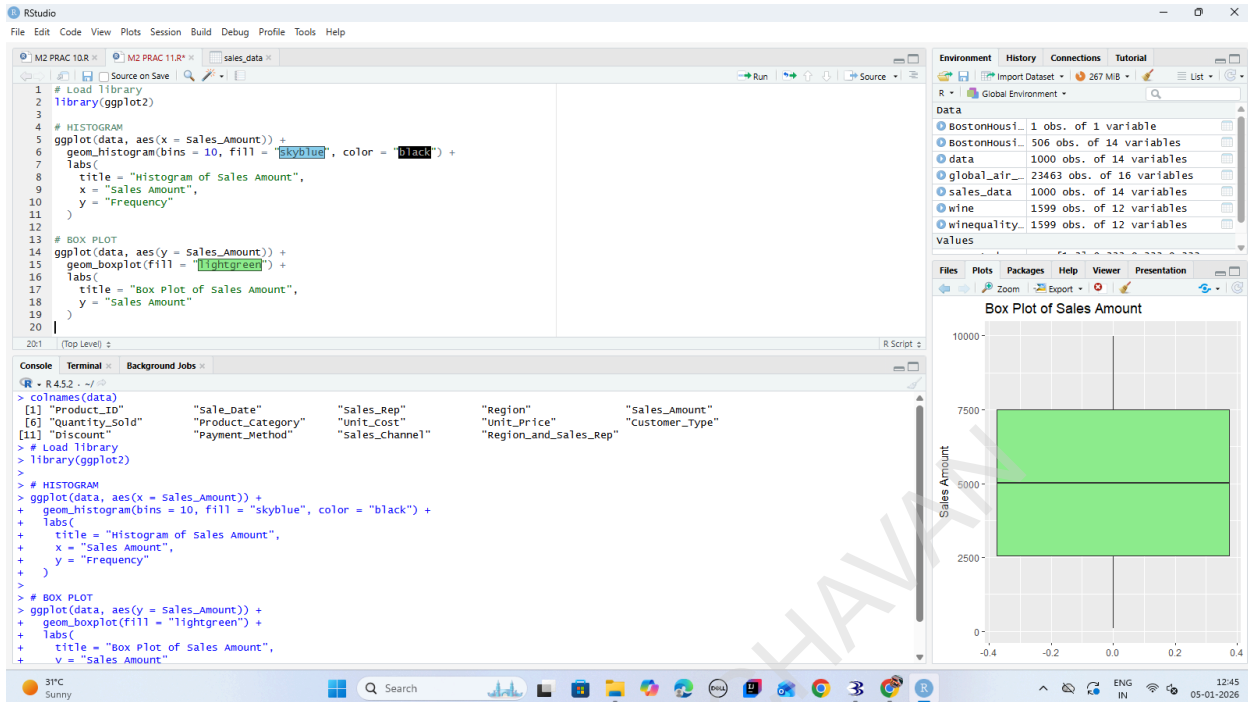
High-Low Chart



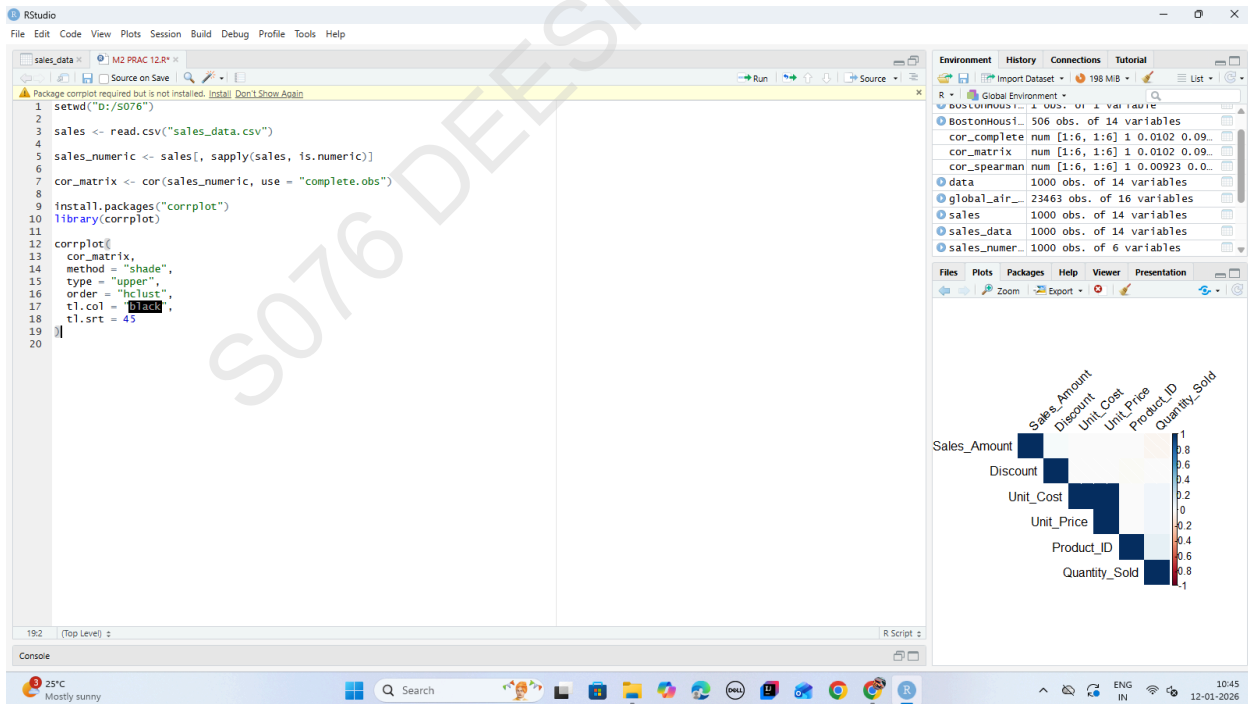
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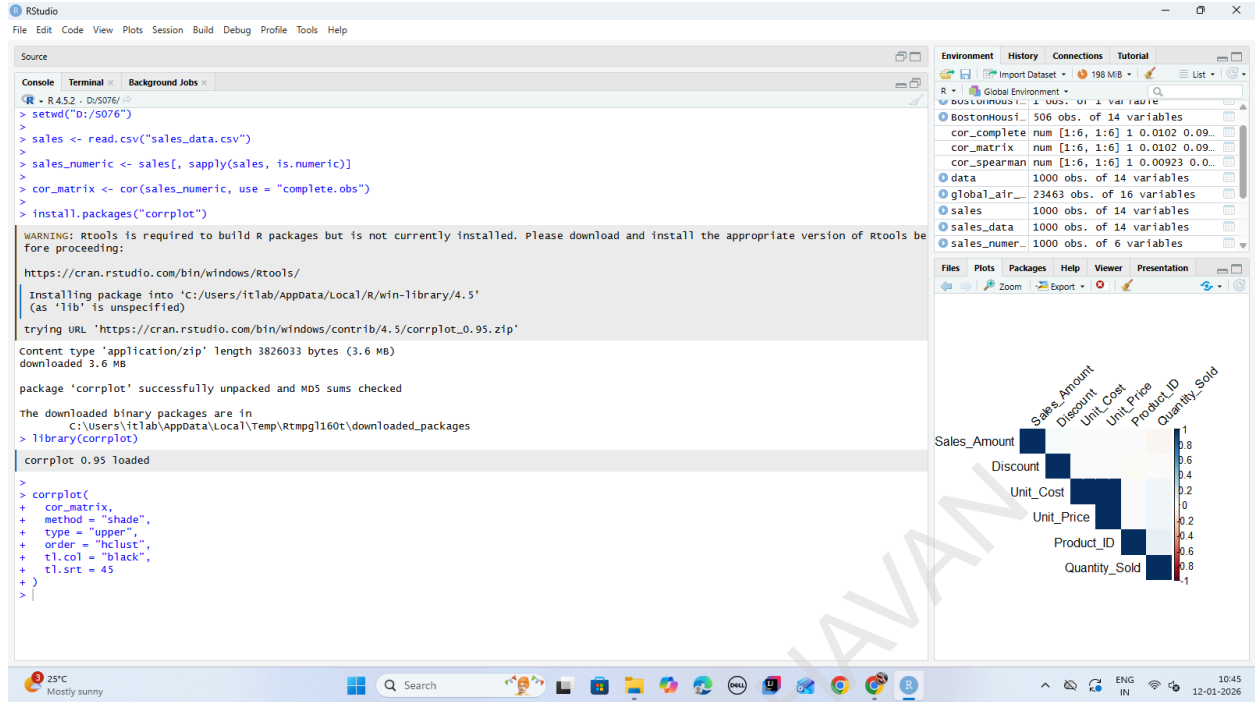
11 Generating histograms and box plots using ggplot2 (R).



12 Generating correlation matrices using cor() (R).



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